PCA Widget ↔ PCA Chart Interaction Diagrams

Interaction flows between the **PCA Widget** (scrollable list of `assembly#haplotype` coordinate keys) and the **PCA Chart** (2D scatter plot with reference background) when the user selects an item in the widget.

The two components are decoupled — they communicate exclusively through `eventBus`. The widget never holds a reference to the chart, and the chart never queries the widget. State convergence happens via three events: `pcaWidget:emphasis`, `pcaWidget:deselect`, and `pcaWidget:absence`.

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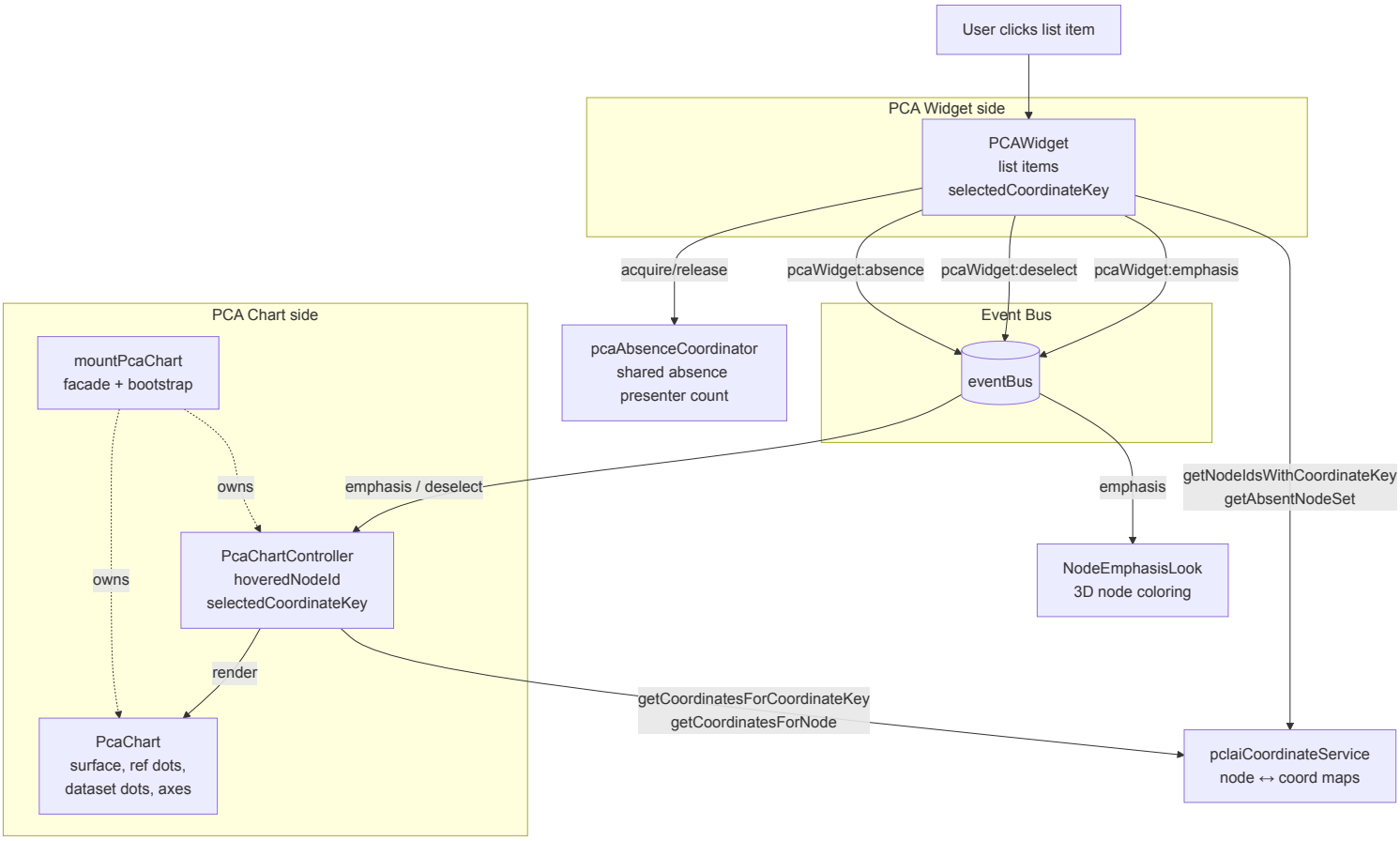
1. Architecture Overview

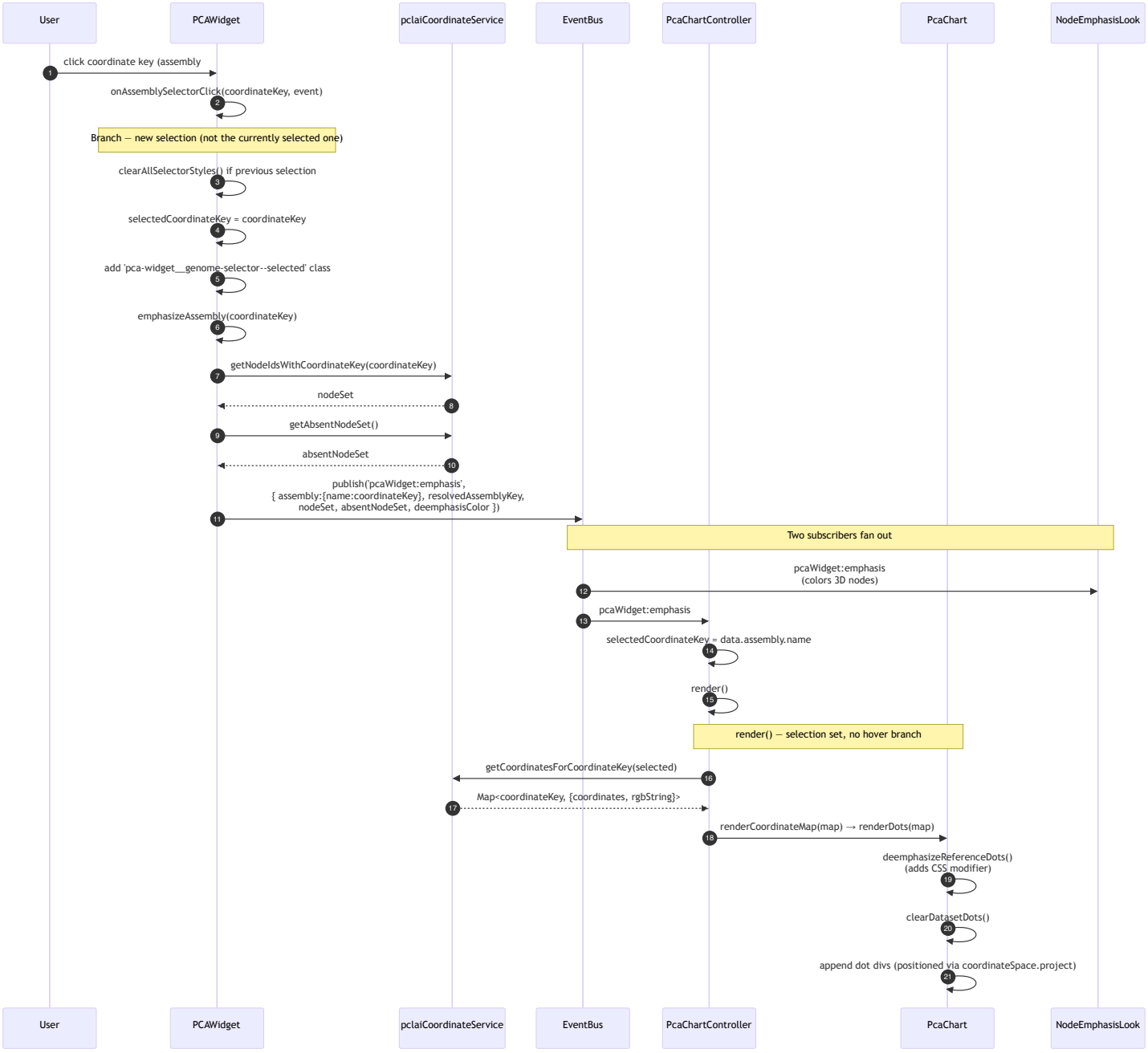
The widget owns the list UI and the user's current selection. The chart owns the surface (background image, reference dots, axes, dataset dots) and renders as a pure function of `(hoveredNodeId, selectedCoordinateKey)` — both inputs arrive over the event bus.

Component	Role	Owns
PCAWidget	List UI + selection source	`selectedCoordinateKey`, list items, selection-toggle behavior
eventBus	Decoupling layer	Pub/sub between widget and chart
PcaChartController	Chart state machine	`hoveredNodeId`, `selectedCoordinateKey`; calls `render()` on every event
PcaChart	View	Surface, reference dots, dataset dots, axes; `renderDots`, `clearChart`, `exportToSvg`
mountPcaChart	Bootstrap facade	Wires controller↔chart, owns reference data, card chrome, button
pclaiCoordinateService	Data source (singleton)	Coordinate maps keyed by node id and by coordinate key
pcaAbsenceCoordinator	Shared absence-mode arbiter	Counts presenters (`pcaWidget`, `pcaChart`) requesting "absent" emphasis



2. Selection — User Clicks a Coordinate Key in the Widget

The most common path. Widget toggles selection state, fires `pcaWidget:emphasis`, controller updates its state and calls `render()`, chart paints dataset dots over the (deemphasized) reference layer.



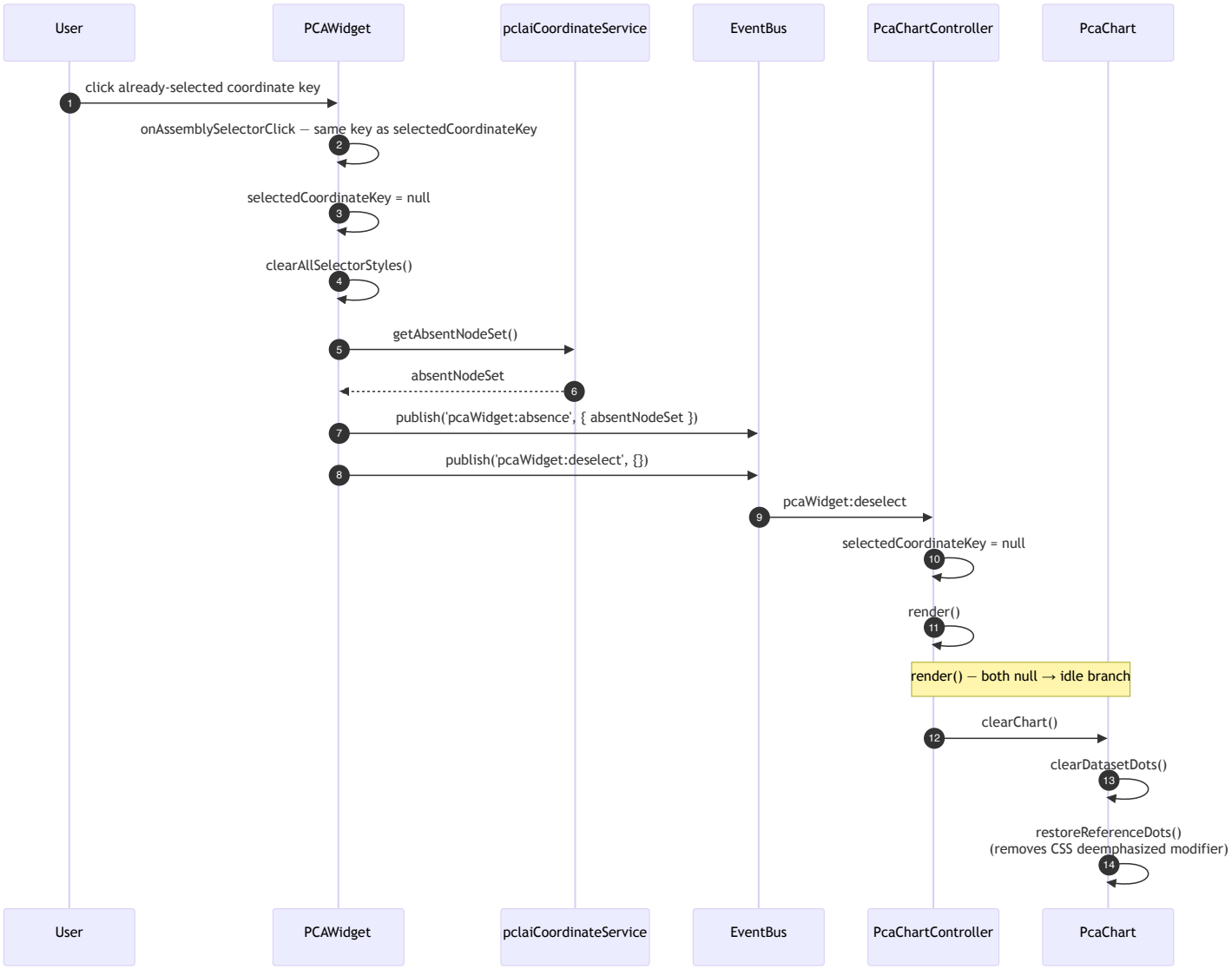
Event Payload — `pcaWidget:emphasis`

Field	Type	Notes
`assembly.name`	string	The coordinate key (`assembly#haplotype`) — controller reads this
`resolvedAssemblyKey`	string	3-part assembly-walk key for downstream consumers (NodeEmphasisLook)
`nodeSet`	Set<string>	Node ids that contain this coordinate key

`absentNodeSet`	Set<string>	Node ids absent from any pclai data
`deemphasisColor`	string	`#aaaaaa` — applied to non-emphasized nodes

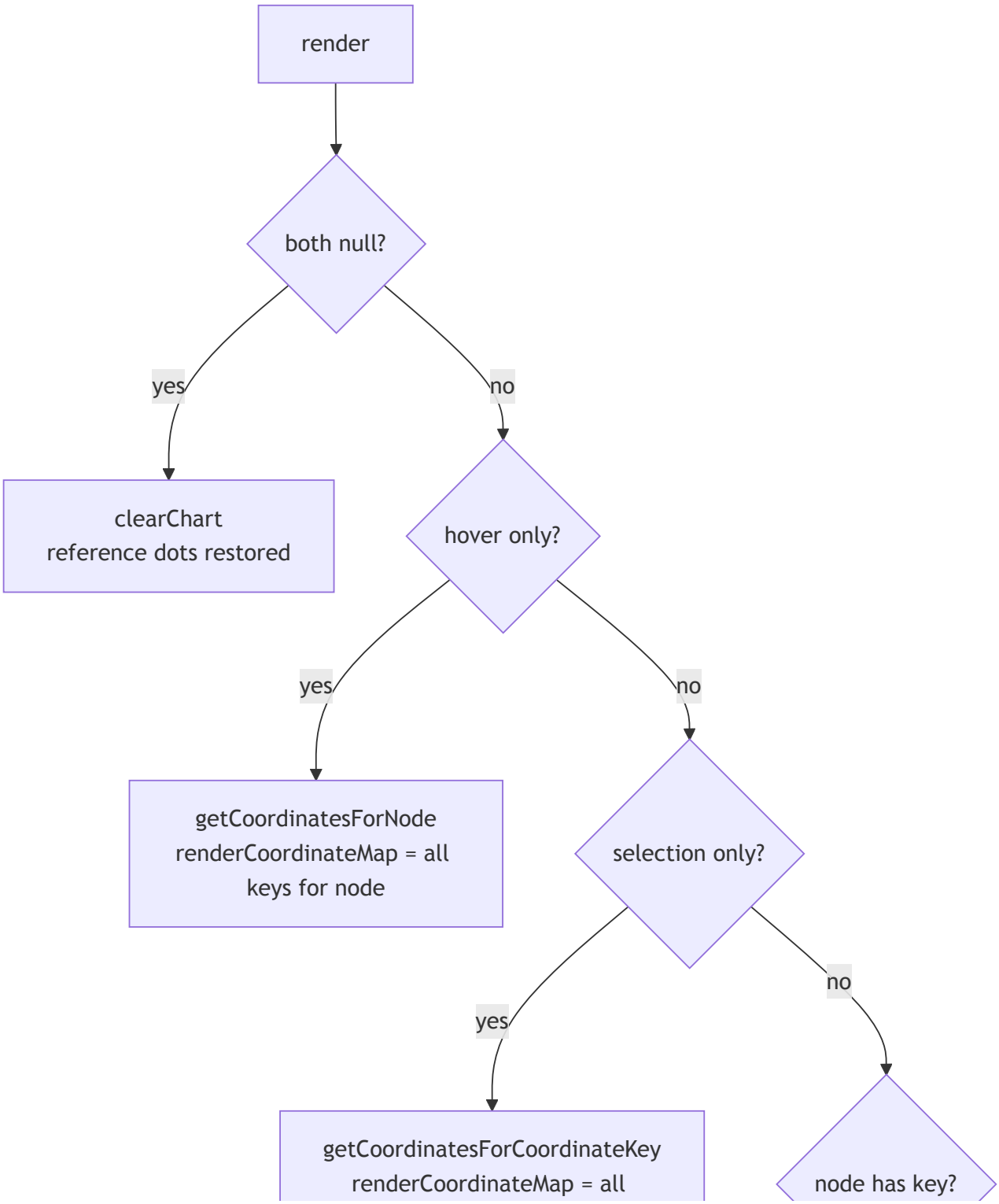
3. Deselection — User Clicks the Currently Selected Item

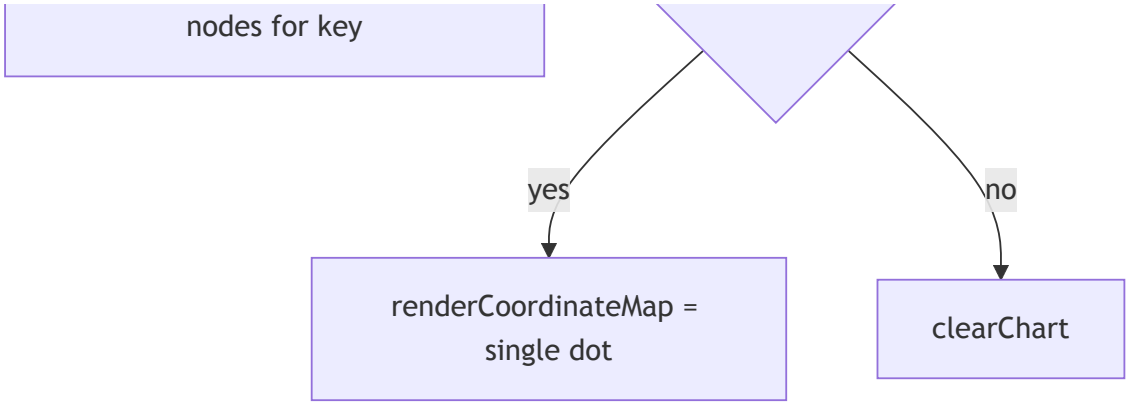
Click toggles. Same item twice → return to idle. Widget clears its selection and fires `pcaWidget:deselect`; controller clears its `selectedCoordinateKey` and `render()` falls into the idle branch (`clearChart()` → reference dots restored to full color).



4. Render State Machine — The Heart of It

The chart is a pure function of `(hoveredNodeId, selectedCoordinateKey)`. Every event handler updates one of those two and calls `render()`. The widget contributes only `selectedCoordinateKey`; `hoveredNodeId` comes from `lineIntersection` / `clearIntersection` events fired by the 3D raycaster (out of scope for this document, but listed for completeness).


render
both null?
yes
no
clearChart
reference dots restored
hover only?
yes
no
getCoordinatesForNode
renderCoordinateMap = all
keys for node
selection only?
yes
no
getCoordinatesForCoordinateKey
renderCoordinateMap = all
node has key?



State	`hoveredNodeId`	`selectedCoordinateKey`	What renders
Idle	null	null	Full-color reference dots only
Hover	set	null	All coord keys for hovered node (over deemphasized reference)
Selection (this doc)	null	set	All nodes for selected key (over deemphasized reference)
Hover ∩ Selection	set	set	Single dot for `(node, key)` if present, else idle

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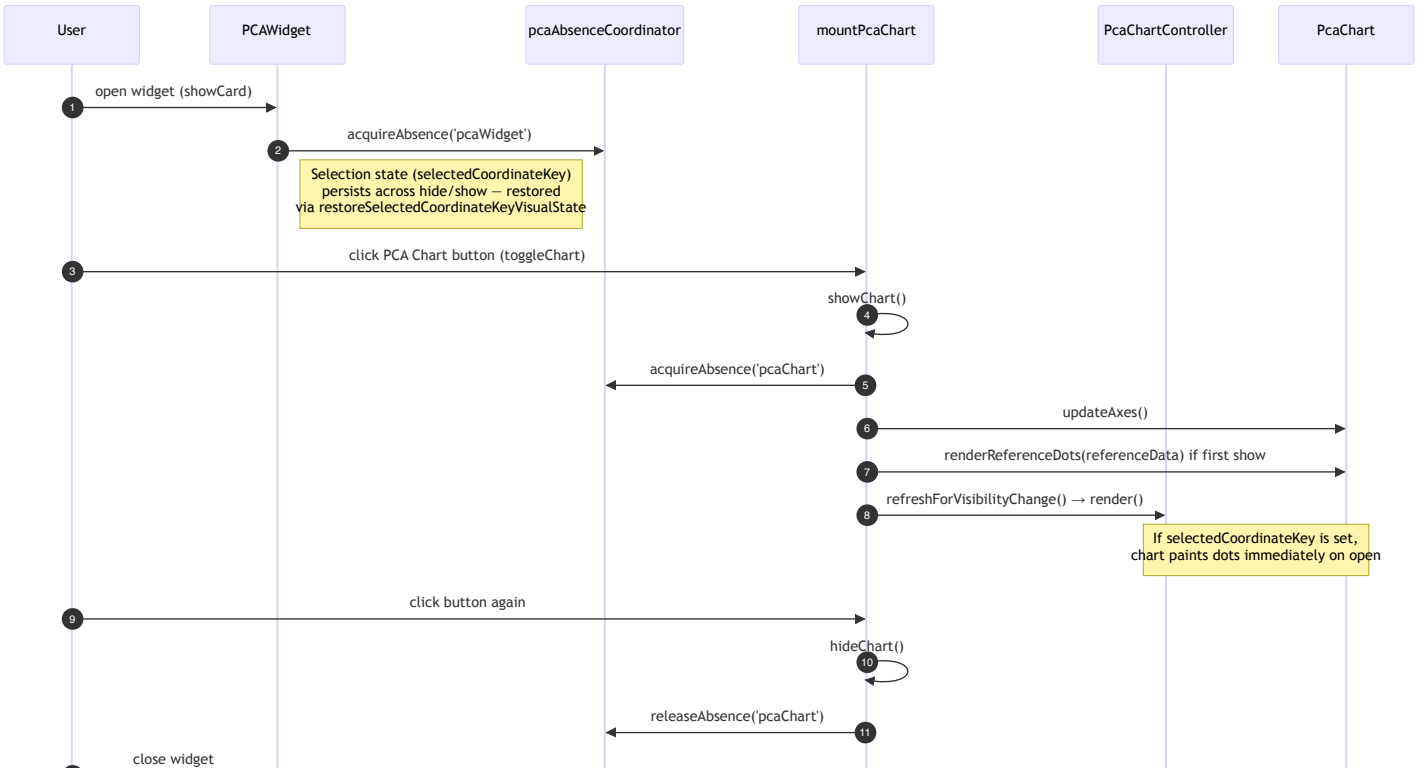
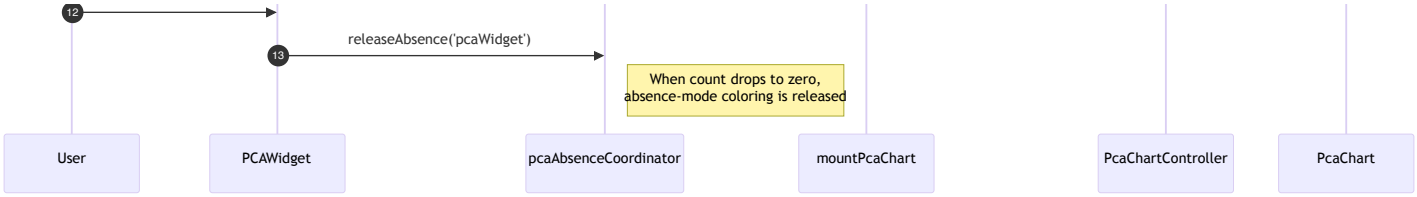
5. Visibility — Widget Card Open/Close vs Chart Card Open/Close

Widget and chart cards are independently shown/hidden. They share the **absence coordinator** (`pcaAbsenceCoordinator`) to track who is requesting absence-mode coloring of nodes that have no pclai data.





Key Points

Aspect	Detail
Decoupling	Widget never references chart objects. All cross-component state flows through `eventBus`.
Selection persistence	Closing the widget does not clear `selectedCoordinateKey` (controller-side). Closing the chart does not affect widget state. Reopening either restores.
Reference dots — first paint	`renderReferenceDots` runs the first time the chart is shown after `initializeGlobalBoundingBox` resolves. Reference data is fetched once at mount and cached.
Background image	Currently a CSS background on the chart surface (`pca_background_576_flipped.png`). Also inlined as a base64 data URI in `exportToSvg`. The runtime load path and the export load path are independent.
Idle vs selected dot rendering	`renderDots` always calls `deemphasizeReferenceDots()` first; `clearChart` always calls `restoreReferenceDots()`. The CSS modifier is the single source of truth for reference-layer state.

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6. Data Flow Summary

```
User clicks coordinate key in PCA Widget
        |
        ▼
PCAWidget.onAssemblySelectorClick
        |
        ├── (toggle / replace selection)
        ├── pclaiCoordinateService.getNodeIdsWithCoordinateKey(key)
        ├── pclaiCoordinateService.getAbsentNodeSet()
        |
        ▼
eventBus.publish('pcaWidget:emphasis', { assembly:{name}, nodeSet, ... })
        |
        ├── NodeEmphasisLook → 3D scene coloring
        |
        └── PcaChartController.handler
                |
                ├── selectedCoordinateKey = data.assembly.name
                ▼
            PcaChartController.render()
                |
                ├── pclaiCoordinateService.getCoordinatesForCoordinateKey(selected)
                |
```

```
            ▼
        PcaChart.renderDots(map)
            |
            ├── deemphasizeReferenceDots()  (CSS modifier on container)
            ├── clearDatasetDots()
            └── append dot divs at coordinateSpace.project(x, y)
```

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7. Key API Used Across the Boundary

From	To	Mechanism	When
PCAWidget	NodeEmphasisLook	`eventBus.publish('pcaWidget:emphasis', ...)`	New selection
PCAWidget	PcaChartController	`eventBus.publish('pcaWidget:emphasis', ...)`	New selection
PCAWidget	PcaChartController	`eventBus.publish('pcaWidget:deselect', {})`	Click on selected toggles off
PCAWidget	NodeEmphasisLook	`eventBus.publish('pcaWidget:absence', { absentNodeSet })`	Selection cleared / replaced
PCAWidget	pclaiCoordinateService	direct call: `getNodeIdsWithCoordinateKey`, `getAbsentNodeSet`	On click
PcaChartController	pclaiCoordinateService	direct call: `getCoordinatesForCoordinateKey`, `getCoordinatesForNode`	Inside `render()`
mountPcaChart	PcaChart	direct method: `renderDots`, `clearChart`, `renderReferenceDots`, `updateAxes`, `exportToSvg`	Through controller's chartDelegate or directly on show

Methods That Are NOT Crossed

- The PCA Widget never calls anything on `PcaChart` or `PcaChartController` directly.
- The PCA Chart never reads `PCAWidget.selectedCoordinateKey` directly — it owns its own copy in the controller, kept in sync via events.